GridPowerAgent: A Grid-Aware LLM Agent for Power System Event Understanding and Tool-Orchestrated Decision Support

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Abstract—**Large language models (LLMs) could act as a coordination layer between power-grid information, engineering analysis tools, and human operators—provided they read operating states accurately, select appropriate tools, and avoid hallucinated advice. We present GridPowerAgent, an LLM agent that observes simulated power-system states, retrieves operating procedures, and orchestrates power-flow, contingency, and optimal-power-flow tools, evaluated on a seeded, fully regenerable corpus: 16,000 operating points and 15,000 scenarios across the IEEE 14/39/118 bus systems, spanning ten disturbance classes with power-flow ground truth, noisy measurements, state estimates, and rule-based labels for retrieval and tool supervision. In a paired 140-scenario $\times$ 4-configuration pilot, a small quantized local model and a lightweight API model diagnose 105–108/140 and 109/140 scenarios respectively, hallucinate on at most 2% of responses, and differ mainly in tool-selection style: the API model defaults to power flow in 52% of its picks, while the local model names scenario-required tools more often under a strict metric. The evaluation also surfaces a result of independent methodological interest for grid-LLM benchmarks: outcome-labeled disturbance classes (e.g., undervoltage injected through line outages) are answered by their cause mechanism, quantifying how mixed-axis label taxonomies penalize causally correct answers and motivating an axis-explicit corpus revision. All results carry exact denominators; findings are observations about this corpus at pilot scale, with a powered evaluation as the next step.**

Index Terms—**power system operation, large language models, LLM agent, tool orchestration, retrieval-augmented generation, state estimation**

I. Introduction

Power-system operators are surrounded by analytical tools—power flow, state estimation, contingency analysis, optimal power flow—yet the work of interpreting their outputs, relating them to operating procedures, and deciding what to do next remains manual and expertise-bound. Large language models (LLMs) are candidates for an *intelligent coordination layer* in this workflow: not replacing conventional analysis, but reading grid states, retrieving the right procedures, invoking the right tools, and explaining the results [1], [2]. Recent agentic systems move in this direction [3], [4], but published evaluations rarely state what the labels are, where they come from, or how the scoring could be reproduced.

This paper presents GridPowerAgent and its evaluation under five research questions: (RQ1) how accurately can an LLM identify normal, abnormal, and critical operating conditions; (RQ2) can it determine which engineering tool a given event requires; (RQ3) can it construct valid tool calls; (RQ4) does grid-specific retrieval improve operational reasoning; and (RQ5) does tool grounding reduce hallucination. RQ3—constructing valid tool calls—requires tool execution, which the pilot does not perform; it is deferred to the powered sweep. This pilot answers RQ1, RQ2, RQ4, and RQ5. We answer them on a corpus built for the purpose: a seeded, headless-resumable pipeline that materializes 16,000 operating points and 15,000 scenarios across the IEEE 14, 39, and 118 bus systems—ten disturbance classes (E0–E9), each scenario carrying pre/post power-flow truth, noisy measurements, state estimates, severity, and rule-based reference labels for retrieval and tool supervision—together with a FAISS knowledge base of operating procedures and four validated physics tools. Four agent configurations (E1 LLM-only, E2 +RAG, E3 +Tools, E4 Full) instantiate the ablation.

The evaluation compares two deployment tiers under identical paired prompts—a 4-bit-quantized small model served locally, and a lightweight API model—so that conclusions do not depend on one provider. All accuracy figures carry exact denominators and Wilson confidence intervals; statistical power is stated alongside every null result; and one finding receives particular attention: both models systematically fail on two outcome-labeled disturbance classes, for a reason we trace to the label taxonomy itself rather than to either model.

Contributions: (i) a grid-aware agent that couples grid-state observation, procedure retrieval, and validated physics tools; (ii) a seeded, hashed, fully regenerable 16k/15k/9.5M-measurement corpus with rule-based tool supervision and a quantified label-noise bound under estimation uncertainty; (iii) a dual-definition tool-scoring protocol that exposes and removes a degenerate answer strategy; (iv) a paired, exactly-denominated pilot across two deployment tiers with an identified label-axis failure mode; and (v) a per-check validation disclosure including unresolved IEEE-39 islanding cases with their scenario identifiers shipped alongside the corpus.

II. Related Work

LLMs have been explored for power-system analysis assistance, dispatch, and contingency response [1], [2], including agentic orchestrators [3], [4] and dispatch benchmarks [5]. Published evaluations seldom disclose how reference labels are produced; our corpus couples per-scenario power-flow truth, noisy measurements, state estimates, and rule-based tool supervision across three IEEE sizes under one seeded pipeline, and scores against it with exact denominators. Foundational RAG [6] and tool-augmented LLMs [7], [8], [9] motivate

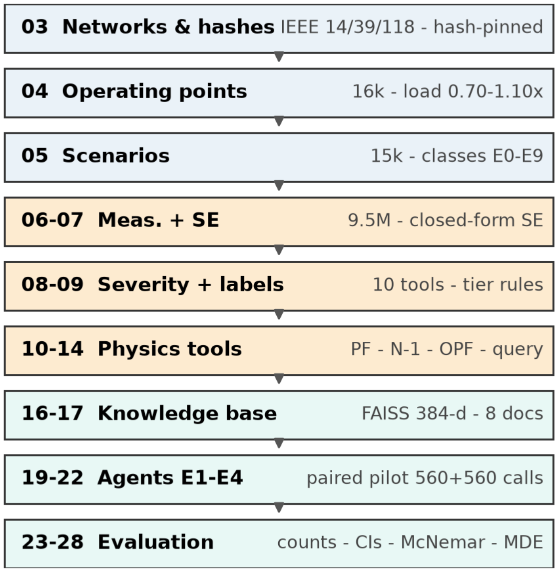

Fig. 1. Pipeline (Stages 03–28).

the agent design. State estimation and measurement modeling follow classical formulations [10], [11]; contingency (N–1) and AC-OPF are standard tools [12], [13], [14]; calibration via ECE [15]; hallucination taxonomy [16]. Synthetic scenarios inherit the spirit of grid-ML benchmarks [17], [18]; pandapower [19] supplies AC power flow; FAISS [20] with sentence-transformers [21] supplies retrieval. On quantization: 4-bit precision is near-optimal for inference scaling [22] and post-training quantization to 3–4 bits is standard [23], though it measurably degrades small models—a conservative bias in our local deployment (Sec. VIII).

III. Methodology

Fig. 1 overviews the pipeline. Stages 03–05 build networks, operating points, and scenarios; 06–09 synthesize measurements, state estimates, severity, and reference labels; 10–14 expose physics tools; 16–17 build the knowledge base; 19–22 run the agent ablations; 23–28 evaluate and render figures. Heavy stages are idempotent and headless-resumable (completed artifacts skipped by row count), materializing the full corpus in $\sim$30 min.

Agent workflow. The agent follows an observe–diagnose–retrieve–plan–execute–interpret loop: it receives the post-event structured grid state (voltage magnitudes, branch loadings, outages, storage state of charge); optionally receives the top-k retrieved operating procedures (E2/E4) and a tool manifest (E3/E4); and must return the disturbance class, a tool selection, and a recommendation with confidence, as JSON. The harness scores the response against the rule-based reference labels; it does not execute tools on the model's behalf in this pilot.

Networks and scenarios. Networks use pandapower IEEE 14/39/118 cases with tuned thermal limits to make compound events observable (14: 3% line / 4% transformer; 118: 6%; 39: nameplate). Limits are disclosed per system and deliberately *not* comparable across systems (Sec. VIII). Operating points sweep load 0.70–1.10$\times$ with $\pm 5\%$ bus-local noise, renewable fractions, and storage state of charge 0.15–0.85 (20 MW/40 MWh at bus 9 on IEEE-14). Topology hashes pin the networks (IEEE-14 `2580e77e`, IEEE-39 `8ded83`, IEEE-118 `c0d6ab`). Ten classes are injected: E0 Normal; E1 load surge, E2 load drop, E3 line outage, E4 generator outage, E5 renewable ramp (cause classes); E6 undervoltage, E7 overvoltage, E8 thermal overload (outcome classes, physically iterated to target severity); E9 compound. The cause/outcome split of E6–E8 matters for the evaluation and is analyzed in Sec. VII-C.

TABLE I
Corpus. Validation is reported per system, not as a single global certificate.

Case	OPs	Scen. (per class)	Meas.	Auto checks	Worst V
14 (ref)	4k	3k (300)	361k	21/21	0.89 pu
39	5k	5k (500)	1.50M	**20/21**	0.89 pu
118	7k	7k (700)	7.68M	21/21	0.89 pu
Total	16k	15k	9.54M	—	—

IEEE-14 limits are artificial (3%/4%) to expose E8; IEEE-39 is at nameplate. The single failing check (s5_no_nan on IEEE-39) covers 41 scenarios with NaN post-voltages from two islanding outages; their identifiers and causes ship with the corpus (case39_nan_scenarios.csv).

IV. Corpus, Simulation and Validation

Table I summarizes the corpus. Scenarios inject the ten classes with 300/500/700 per class on 14/39/118; replay over 40 draws confirms exact determinism.

Measurements model $\sigma_V = 0.003$ pu and $\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05)$ MVA computed from the noise-free power-flow solution (Eqs. 1–2). State estimates are a closed-form noise-aware approximation ($\hat{v} = v + \epsilon$, $\epsilon \sim \mathcal{N}(0, 0.0007^2)$), *not* full iterative WLS with assembled Jacobians; replacement with `pandapower.estimation.estimate` is future work and we do not claim paper-grade estimator fidelity.

$$z = h(x) + \epsilon, \quad \epsilon_V \sim \mathcal{N}(0, 0.003^2) \tag{1}$$

$$\sigma_P = \max(0.0075|S_{\text{true}}|, 0.05),\ W = \operatorname{diag}(\sigma^{-2}) \tag{2}$$

$$\hat{x} = \arg\min_x \, (z - h(x))^\top W (z - h(x)) \tag{3}$$

Severity score. The illustrative severity used for reference labels is

$$S = 0.6\ \min\Big(1, \tfrac{\max(0, |V-1|-0.02)}{0.06}\Big) + 0.4\ \min\Big(1, \tfrac{\max(0, \ell-L)}{10}\Big), \tag{4}$$

with boundaries 0.0263/0.0526/0.1053 selected by grid search over five candidate sets *on this corpus*—a circularity we return to in Sec. VIII.

Label noise under estimation uncertainty. Reference labels gate two tools on severity. Recomputing severity with estimated bus voltages yields rank agreement $\rho = 0.764$

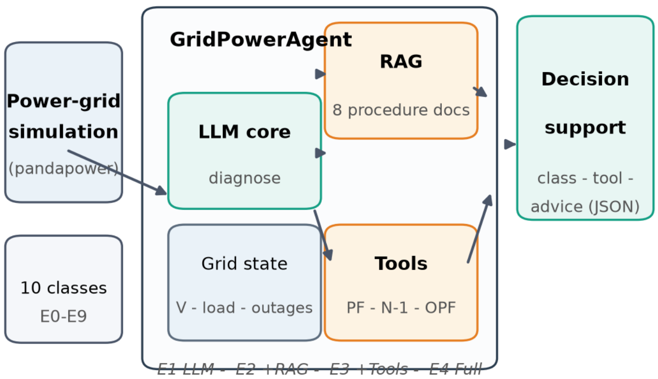

Fig. 2. Agent architecture.

TABLE II
EXPERIMENTAL ACCOUNTING (EXACT DENOMINATORS EVERYWHERE).

Item	Setting
Scenario sample	seeded draws from 3,000 IEEE-14 scenarios; identical set for both models
Configurations	E1 / E2 / E3 / E4 (paired, identical prompts)
Calls	140 × 4 = 560 per model; 1,120 total
Models	lightweight API model; small quantized local model (same prompts)
Harness oracle	rule-based, N=600/config (integration test only)
Statistics	exact counts; Wilson 95% CI; exact McNemar; bootstrap CI; MDE

(14), 0.968 (39), 0.823 (118)—the voltage term sits near its deadband for most scenarios, so noise reorders ranks without crossing thresholds. The operationally relevant quantity is accepted-set membership: scoring-relevant label flips affect **0/30,000** scenario–tool judgments on IEEE-14 (the pilot system) and **31/150,000** (0.021%) corpus-wide. The binary violation detector reconciles 99.23%/99.54%/99.67% on 14/39/118.

V. Retrieval, Tools and Agent Design

The knowledge base chunks eight operational documents (thermal/voltage limits, contingency procedure, storage/equipment, topology) into a FAISS index (384-d sentence-transformer embeddings), retrieving $k = 3$ chunks with citations. Recall@1 is 100% on held probes; with eight documents this is a plumbing check, not evidence of retrieval quality—hard-negative evaluation is future work. Four physics tools (power flow, grid queries, contingency N–1, OPF) are validated 21/21 including 15/15 contingency convergences.

VI. Evaluation Protocol

Models. Two deployment tiers of the same prompt contract are compared: a lightweight API model (Gemini 3.5 Flash Lite, temperature 0, ≤512 output tokens, standard rate-limit retry with checkpointed resume) and a small 4-bit-quantized open-weight model (Gemma 4 E4B-it [24]) served locally from a GGUF build via llama.cpp [25] (temperature 0, ≤1,024 tokens; its chat template emits a visible reasoning channel, and parsing reads `content` with fallback to the reasoning text). *The API tier is a lightweight, latency-optimized model*: the comparison characterizes a representative local deployment against a budget API tier, not against frontier API models. A deterministic rule-based *oracle* (not a model) validates harness

TABLE III
PAIRED PILOT, EXACT COUNTS WITH WILSON 95% CIS. TOOL COLUMN: STRICT-SPECIFIC METRIC (DEFINED IN THE EVALUATION PROTOCOL).

	Diagnosis		Tool (strict)	
Cfg	API (n=140)	Local (n=140)	API	Local
E1	109/140 [70, 84]	107/140 [69, 83]	53/140 [30, 46]	64/140 [38, 54]
E2	109/140 [70, 84]	108/140 [70, 83]	50/140 [28, 44]	63/140 [37, 53]
E3	109/140 [70, 84]	105/140 [67, 81]	36/140 [19, 34]	64/140 [38, 54]
E4	109/140 [70, 84]	107/140 [69, 83]	35/140 [19, 33]	64/140 [38, 54]

Latency: API 1.06–1.16 s (round-trip); Local 42–70 s (visible-thinking decode).
Hallucinated rows (any of 6 tags): API 2/1/1/0 (E1–E4); Local 1/1/2/2 (E1–E4).
Deterministic baselines on the same sample: majority-class 20%, random 12%±7%.

plumbing at N=600/config; statistics on it test code, not intelligence.

Metrics. Diagnosis correctness is exact class match. *Tool selection is reported under two definitions.* (a) *Accepted-set*: the stated tool is required or strongly appropriate. This definition is degenerate—power flow lies in every scenario's accepted set, so always answering power flow scores 100%—and we treat it as an upper bound only. (b) *Strict-specific*: the stated tool must be *required* for the scenario, and power flow counts only when it is the sole required tool. Bare `grid_query` answers are scored non-specific. Hallucination is response-level, any of six tags (H-NUM/TOP/EQP/PHY/TOOL/ACT), assigned by an automated rule-based judge; rates are judge agreement, not ground truth (Sec. VIII).

Power. At n=140 paired scenarios with the observed discordance rates, the minimum detectable difference (two-sided, α=0.05, power 0.80) is E1: 2.8, E2: 2.0, E3: 3.9, E4: 2.8 pp. Null results in this paper therefore mean "no difference above the per-configuration MDE was detectable," not "the models are equivalent"; a 600-scenario-per-configuration sweep is scripted as the powered follow-up.

VII. Results

A. RQ1/RQ4: Event Diagnosis and the Effect of Retrieval

Table III reports exact counts with Wilson 95% CIs. The API model diagnoses 109/140 (77.9%) in *every* configuration; the local model 105–108/140 (75.0–77.1%). All diagnosis discordances are one-directional (2/1/4/2 across E1–E4: wherever the models disagree, the API model is right), and exact McNemar tests detect no significant difference (Table IV)—as expected at this power: the minimum detectable difference is E1: 2.8, E2: 2.0, E3: 3.9, E4: 2.8 pp. Configuration additions (retrieved procedures, tool manifests) changed not a single diagnosis for either model: the structured grid state in the prompt already carries the discriminating information, which answers RQ4 negatively *for diagnosis* at this corpus scale.

B. RQ2: Tool Selection and the Default-Answer Bias

Tool selection is where the deployment tiers differ most—in *style* before *accuracy*. Under the permissive accepted-set metric both models score 100%: the metric is degenerate, since always answering power flow achieves it by construction (random tool choice already scores 53%). The API model exercises exactly this strategy: 52% of its stated tools

TABLE IV
EXPLORATORY PAIRED ANALYSIS (LOCAL − API DIAGNOSIS, PERCENTAGE POINTS). BOOTSTRAP 95% CI, 20K RESAMPLES. MARGINS WERE NOT PRE-REGISTERED; NO PASS/FAIL IS CLAIMED.

Cfg	Pairs	Diff (pp)	95% CI	McNemar p
E1	140	-1.4	[-3.6, 0.0]	0.50
E2	140	-0.7	[-2.1, 0.0]	1.00
E3	140	-2.9	[-5.7, -0.7]	0.12
E4	140	-1.4	[-3.6, 0.0]	0.50

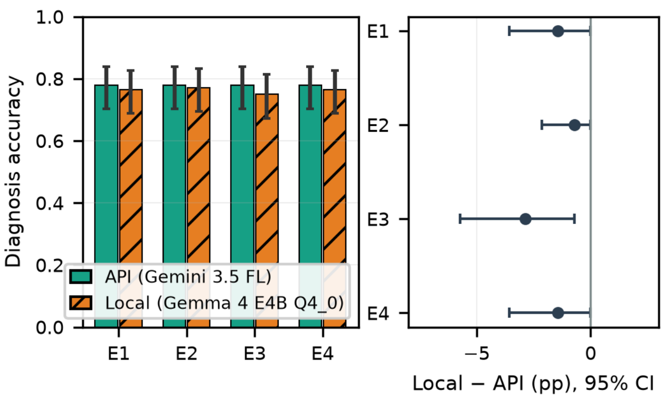

Fig. 3. (a) Diagnosis accuracy, local vs. API model, Wilson 95% CIs. (b) Paired diagnosis difference with bootstrap 95% CIs against reference margins (−10pp, −5pp).

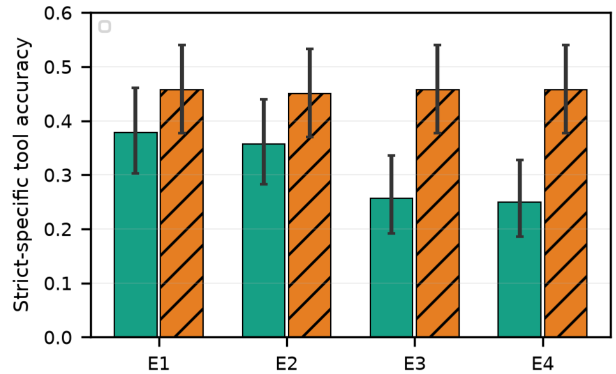

Fig. 4. Strict-specific tool-selection accuracy (reference required set). Exact counts in Table III.

TABLE V
E4 PER-CLASS, POOLED MODELS (RAW COUNTS; NO PERCENTAGES).

Class	Obs.	Diag.	Tool (strict)
E0	10	10/10	0/10
E1	26	26/26	0/26
E2	20	20/20	0/20
E3	28	28/28	28/28
E4	32	32/32	26/32
E5	38	38/38	0/38
E6	36	0/36	0/36
E7	22	22/22	0/22
E8	26	0/26	19/26
E9	42	40/42	26/42

Obs. = scenarios × models. Classes absent from the pilot sample: none.

are power flow. Under the strict-specific metric, the local model names a scenario-*required* tool in 63/140–64/140 of cases (flat across configurations) versus 35/140–53/140 for the API model. The conditioning behind these rates sharpens the default-answer-bias reading. When the API model states power flow (291 responses), that tool is actually required for the scenario in only 35% of cases—the rest are defaults to the universally accepted answer. The local model states power flow rarely (49 responses, 24% of them required); its dominant choices are contingency (327 responses, 78% required—contingency is required precisely on outage and overload scenarios) and equipment queries. In other words, the two models' strict scores measure how often each model's committed tool matches the scenario's actual requirement, and the gap comes from the API model's default rate, not from superior local tool knowledge; both remain far from ceiling, and the binary metric cannot distinguish well-placed tools from lucky guesses. Per-class structure is analyzed next.

C. A Label-Axis Failure Mode: E6 and E8

Table V shows near-ceiling diagnosis on every cause-labeled class and total collapse on two outcome-labeled classes: E6 (undervoltage) and E8 (thermal overload) are diagnosed correctly in 0 of 36 and 0 of 26 pooled observations. The failure is systematic, not stochastic: across all 144 failed E6 responses, 128 predict E3, 16 predict E4—and across all 104 failed E8 responses, 101 predict E9, 3 predict E3. The logged reasons are explicit: "The injected event is a single transmission line outage (line_6_13), which directly corresponds to class E3" (an E6 scenario); "the injected event description explicitly states a compound mechanism consisting of a transmission line outage and a 20% demand increase" (an E8 scenario).

The mechanism is a mixed-axis label taxonomy. E6/E8 are *outcome* classes—defined by the post-event state—but are *injected* through cause mechanisms (line outages that cause undervoltage; line-outage-plus-load-surge compounds that cause overloads). Both models classify on the cause axis and are therefore *causally correct but label-wrong* on every E6/E8 scenario. This is a property of the benchmark's labeling scheme, not evidence about model capability, and it is the kind of flaw a benchmark must surface about itself: mixed-axis taxonomies silently reward or punish models for answer-style choices unrelated to the construct being measured. Correcting it requires either relabeling E6/E8 as cause events or instructing the taxonomy's outcome primacy explicitly; both are queued for the next corpus revision, and the powered sweep will report both cause-axis and label-axis accuracy.

D. RQ5: Hallucination and the Cost of Local Inference

Any-tag hallucinated rows are rare for both models under the automated judge: 2/1/1/0 (E1–E4) (API) and 1/1/2/2 (E1–E4) (Local) across E1–E4. These rates mean "the rule-based

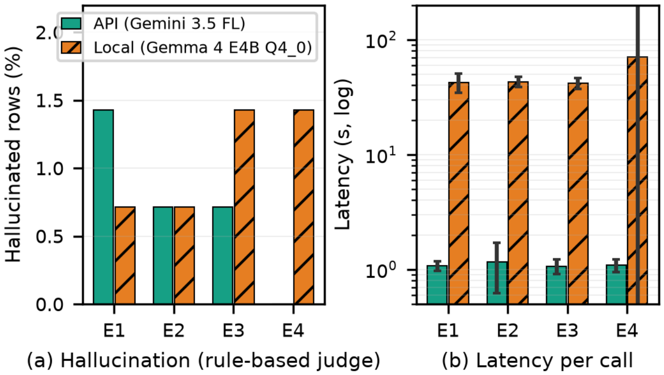

Fig. 5. Response-level any-tag hallucination (automated rule-based judge) and latency (log scale). Exact denominators in Table III.

TABLE VI
LOGGED E4 RESPONSE, SCENARIO IEEE14_SCN_002952 (API MODEL).

Field	Logged content
Answer	“‘json ”event_class”: ”E9”, ”confidence”: 1.0, ”tool”: ”power_flow”, ”reason”: ”The injected description explicitly specifies a compound event involving both a transmission line outage and a load surge.” “‘
Judge	diag correct (E9); tool = power_flow $\in$ accepted set; no H-TOP/H-TOOL flags

judge flagged no row”; they are not expert-annotated ground truth. Latency is the deployment trade-off: local inference averages 42–70 s per call versus 1.06–1.16 s for the API round-trip—roughly a 45× difference—while carrying zero marginal API cost, no rate-limit dependency, and no grid data leaving the premises. For always-on monitoring this may be acceptable; for closed-loop use neither model's latency profile is appropriate (Sec. VIII).

Harness oracle (integration test, not evidence). On the seeded 600-scenario oracle run the E1→E4 ladder moves 55.7%→88.5% diagnosis (McNemar $p = 3.1 \times 10^{-32}$). These statistics validate that the harness can *detect* configuration differences; because the oracle is programmed to respond differently when RAG/tools are present, they carry no evidence about real LLMs.

Qualitative trace (from the raw log). Table VI quotes the actual logged response for scenario IEEE14_SCN_002952 ($V_{\min}$ 0.970 pu, $V_{\max}$ 1.060 pu, peak loading 1.4%, violations 0). The pilot does not execute tool calls; the “tool” field is the model's stated choice.

Calibration. At pilot N with near-saturated per-class correctness, confidence calibration is unidentifiable; the oracle's ECE numbers characterize the confidence simulator, not any model. ECE will be reported on the powered sweep.

VIII. Discussion, Limitations and Ongoing Work

Scope. The evidence covers *one model pair* (a 4-bit small local model; a lightweight API tier) on *one corpus*, one prompt template, temperature 0, single runs. The API tier is deliberately lightweight; results do not speak to frontier API models, and nothing here claims that local models match API models in general. The minimum detectable difference at this sample size is ∼3pp (Sec. VI); the powered 600-scenario sweep is the inferential step.

Label circularity. Disturbances, labels, and prompts derive from the same rule family: the evaluation measures *recovery of the synthetic labeling policy*, not open-ended operator reasoning—and the severity boundaries of Eq. (4) were themselves tuned on this corpus. Required remedies, all future work: expert-reviewed labels (2×80, target $\kappa \geq 0.8$), acceptance of multiple valid tool sequences, blind scoring of final recommendations, an evaluator independent of the label generator, and severity boundaries anchored to an external operating standard.

Taxonomy design. The E6/E8 label-axis failure (Sec. VII-C) is a benchmark-design finding: outcome-labeled classes injected via cause mechanisms make cause-correct answers score zero. The next corpus revision separates the axes explicitly.

Physics fidelity. Thermal limits differ per system by construction, so overload magnitudes are not operationally comparable across systems. The corpus is static: no dynamics, protection, or time-sequential behavior. The severity index is illustrative with a quantified label-noise bound (Sec. IV); the state estimator is a closed-form approximation pending replacement with pandapower's iterative estimator; the 41 IEEE-39 islanding scenarios (IDs and causes shipped in `case39_nan_scenarios.csv`) remain unresolved and are excluded from any downstream use of post-voltage values.

Scoring subjectivity. Hallucination tags come from a single automated rule-based judge; “near-zero hallucination” means “the automated judge flagged no row.” Tool scoring relies on the reference policy; the strict metric narrows but does not eliminate policy-dependence.

Quantization. The local model ran at 4-bit quantization, which measurably degrades small-model fidelity [22], [23]; local results are a conservative lower bound for higher-precision serving.

Safety framing. This is an offline advisory research prototype. No switching validation, protection coordination, authorization, or human-approval interface is implemented or claimed; closed-loop use is out of scope.

Reproducibility and generalization. All generation scripts, seeds, hashes, prompt templates, per-stage validation summaries, and the IEEE-39 NaN identifier list are in the repository; corpora regenerate deterministically from seeds, and an archived (DOI) release is planned before submission. Generalization to the larger networks is *planned, not performed*: no agent ran on IEEE-39/118 in this paper.

Ongoing work. (1) The powered 600-scenario × 4-configuration sweep per model with per-class F1 under both label axes, ECE, multiplicity-corrected paired tests, and prompt/temperature sensitivity; (2) the taxonomy revision separating cause and outcome axes; (3) replacement of the state estimator; (4) expert annotation; (5) measured transfer to IEEE-39/118.

IX. Conclusion

GridPowerAgent couples a seeded, fully regenerable 16k/15k power-system scenario corpus with a grid-aware LLM agent that retrieves operating procedures and orchestrates validated physics tools, and evaluates it under a protocol where every number carries an exact denominator and every null result carries its statistical power. In a paired pilot across two deployment tiers, event diagnosis was statistically indistinguishable between a small local model and a lightweight API model, tool choice differed mainly in default-answer bias, and a systematic failure mode was traced to the benchmark's own label taxonomy rather than to either model. The agent, the corpus, and the protocol—including their disclosed flaws—are the contribution; the powered evaluation they now enable is the next step.

Data Availability: Generation scripts, seeds, hashes, prompt templates, per-stage validation summaries, and the IEEE-39 NaN scenario identifier list are in the repository; corpora regenerate deterministically from seeds. An archived (DOI) release of code and a corpus sample is planned before submission.

AI Assistance Disclosure: Generative AI was used for drafting, coding assistance, and figure generation. All experimental decisions, execution, and analysis were performed by the authors, who reviewed and edited all AI-assisted content.

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